

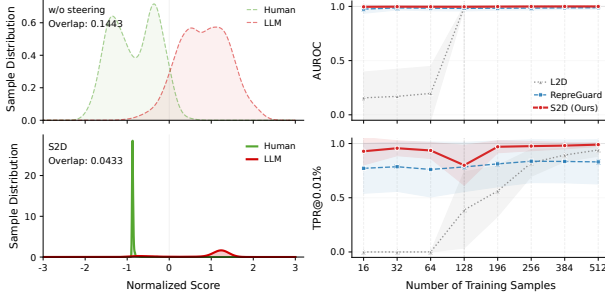


Figure 4: **Detection analysis.** Left: Steering leads to better separability. Right: Detection performance across training sizes. Full results are in Figures 7 and 8 in Appendix.

This dependence is consistent with the design of S2D: the method operates on the hidden representations of a frozen observer and can amplify discriminative structure already present in its latent space, but its effectiveness is limited when human- and LLM-generated representations remain strongly entangled. We therefore do not expect all observer models to perform equally well. In practice, we recommend selecting the observer using a small held-out validation set, with pre-steering representation overlap serving as a useful diagnostic of observer suitability.

Shots of Training Dataset. Finally, we investigate the impact of training set size on detection performance using S2D, ReprGuard, and L2D, since these three methods yielded the best overall results in the preceding experiments. As shown in Figure 4 (right), S2D achieves strong performance even in the low-shot regime, while ReprGuard shows moderate sensitivity, and L2D degrades substantially with limited data. This indicates that S2D can leverage intrinsic differences between human-written and LLM-generated text in the representation space, with training mainly serving to reduce distributional overlap. Detailed results are provided in the Appendix F.3.

5 Discussion

We present S2D, a representation-based framework for detecting LLM-generated text. By injecting a lightweight steering vector into hidden representations, S2D improves class separability and formulates detection as a hypothesis test. The resulting detector provides Type-I error control and a bound on excess Type-II error. Experiments demonstrate strong performance across ID and OOD settings, as well as robustness to adversarial perturbations, paraphrasing, varying text lengths, and different observer models. S2D also achieves a favorable trade-off between detection performance and computational efficiency compared with existing methods.

Several limitations suggest directions for future work. First, S2D is a document-level binary detector and does not localize human- and AI-written segments within mixed-authorship text. Second, a single vMF model may not fully capture latent geometry across domains; mixture models or target-domain adaptation may improve robustness. Finally, more flexible steering mechanisms, such as multiple vectors or non-additive interventions, could better capture representation geometry.

References

- [1] Inyoung Cheong, King Xia, KJ Kevin Feng, Quan Ze Chen, and Amy X Zhang. (A) I am not A lawyer, but...: engaging legal experts towards responsible LLM policies for legal advice. In *Proceedings of the 2024 ACM conference on fairness, accountability, and transparency*, pages 2454–2469, 2024.
- [2] Giorgos Iacovides, Thanos Konstantinidis, Mingxue Xu, and Danilo Mandic. FinLlama: LLM-based financial sentiment analysis for algorithmic trading. In *Proceedings of the 5th ACM International Conference on AI in Finance*, pages 134–141, 2024.
- [3] Dmitry Kobak, Rita González-Márquez, Emőke-Ágnes Horvát, and Jan Lause. Delving into LLM-assisted writing in biomedical publications through excess vocabulary. *Science Advances*, 11(27):eadt3813, 2025.
- [4] Hanchen Su, Wei Luo, Yashar Mehdad, Wei Han, Elaine Liu, Wayne Zhang, Mia Zhao, and Joy Zhang. LLM-friendly knowledge representation for customer support. In *Proceedings of the*

Observer Model	AUROC	TPR@1%	TPR@.01%
Llama-3.1-8B	99.62 ± 0.46	99.05 ± 0.60	97.98 ± 1.37
Mistral-7B-v0.3	68.52 ± 0.94	30.53 ± 5.31	23.28 ± 6.45
GPT-Neo-2.7B	82.38 ± 2.94	32.65 ± 8.91	11.33 ± 5.67
OPT-2.7B	99.86 ± 0.13	98.30 ± 1.79	84.50 ± 8.24
Qwen2.5-7B	99.93 ± 0.13	99.58 ± 0.65	98.65 ± 1.69
Falcon-7B	99.96 ± 0.05	99.40 ± 0.88	98.28 ± 1.83
Falcon-7B-Inst.	99.96 ± 0.05	99.18 ± 1.45	95.90 ± 2.68
Gemma-2-9B	76.54 ± 14.6	19.25 ± 5.37	3.05 ± 2.89
Gemma-2-9B-Inst.	79.09 ± 20.3	37.73 ± 26.6	18.90 ± 14.1

Table 5: **Observer stability.** Detection performance (mean ± std), averaged across four training datasets. Full results are in Table 9 in Appendix.

- 31st International Conference on Computational Linguistics: Industry Track, pages 496–504, 2025.
- [5] Devesh Batra, Conor Hamill, John Hartley, Ramin Okhrati, Dale Seddon, Harvey Miller, Raad Khraishi, and Greig Cowan. A review of LLM agent applications in finance and banking. Available at SSRN 5381584, 2025.
 - [6] Jooyoung Lee, Thai Le, Jinghui Chen, and Dongwon Lee. Do language models plagiarize? In *Proceedings of the ACM Web Conference 2023*, pages 3637–3647, 2023.
 - [7] Aneta Zugecova, Dominik Macko, Ivan Srba, Robert Moro, Jakub Kopál, Katarína Marcinčinová, and Matúš Mesarčík. Evaluation of LLM vulnerabilities to being misused for personalized disinformation generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 780–797, 2025.
 - [8] Jiashu Xu, Mingyu Ma, Fei Wang, Chaowei Xiao, and Muhao Chen. Instructions as backdoors: Backdoor vulnerabilities of instruction tuning for large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 3111–3126, 2024.
 - [9] Yifan Yang, Qiao Jin, Furong Huang, and Zhiyong Lu. Adversarial prompt and fine-tuning attacks threaten medical large language models. *Nature Communications*, 16(1):9011, 2025.
 - [10] Eric Mitchell, Yoonho Lee, Alexander Khazatsky, Christopher D Manning, and Chelsea Finn. DetectGPT: Zero-shot machine-generated text detection using probability curvature. In *International conference on machine learning*, pages 24950–24962. PMLR, 2023.
 - [11] John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. A watermark for large language models. In *International Conference on Machine Learning*, pages 17061–17084. PMLR, 2023.
 - [12] Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, and Weijie J Su. A statistical framework of watermarks for large language models: Pivot, detection efficiency and optimal rules. *The Annals of Statistics*, 53(1):322–351, 2025.
 - [13] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.
 - [14] Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, Hadas Kotek, and Yonatan Belinkov. LLMs know more than they show: On the intrinsic representation of LLM hallucinations. *arXiv preprint arXiv:2410.02707*, 2024.
 - [15] Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. *Advances in Neural Information Processing Systems*, 36:41451–41530, 2023.
 - [16] Oscar Skea, Md Rifat Arefin, Dan Zhao, Niket Patel, Jalal Naghiyev, Yann LeCun, and Ravid Shwartz-Ziv. Layer by layer: Uncovering hidden representations in language models. *arXiv preprint arXiv:2502.02013*, 2025.
 - [17] Seongheon Park, Xuefeng Du, Min-Hsuan Yeh, Haobo Wang, and Yixuan Li. Steer LLM Latents for Hallucination Detection. In *International Conference on Machine Learning*, pages 47971–47990. PMLR, 2025.
 - [18] Jingtao Sun and Zhanglong Lv. Zero-shot detection of LLM-generated text via text reorder. *Neurocomputing*, 631:129829, 2025.
 - [19] Jinyan Su, Terry Zhuo, Di Wang, and Preslav Nakov. DetectLLM: Leveraging log rank information for zero-shot detection of machine-generated text. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 12395–12412, 2023.
 - [20] Guangsheng Bao, Yanbin Zhao, Zhiyang Teng, Linyi Yang, and Yue Zhang. Fast-DetectGPT: Efficient Zero-Shot Detection of Machine-Generated Text via Conditional Probability Curvature. In *The Twelfth International Conference on Learning Representations*, 2024.

- [21] Abhimanyu Hans, Avi Schwarzschild, Valeriia Cherepanova, Hamid Kazemi, Aniruddha Saha, Micah Goldblum, Jonas Geiping, and Tom Goldstein. Spotting LLMs with binoculars: Zero-shot detection of machine-generated text. *arXiv preprint arXiv:2401.12070*, 2024.
- [22] Chengzhi Mao, Carl Vondrick, Hao Wang, and Junfeng Yang. Raidar: generative AI detection via rewriting. *arXiv preprint arXiv:2401.12970*, 2024.
- [23] Hongyi Zhou, Jin Zhu, Erhan Xu, Kai Ye, Ying Yang, and Chengchun Shi. Learn-to-Distance: Distance Learning for Detecting LLM-Generated Text. *arXiv preprint arXiv:2601.21895*, 2026.
- [24] Yifei Huang, Jiuxin Cao, Hanyu Luo, Xin Guan, and Bo Liu. Magret: Machine-generated text detection with rewritten texts. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 8336–8346, 2025.
- [25] Irene Solaiman, Miles Brundage, Jack Clark, Amanda Askell, Ariel Herbert-Voss, Jeff Wu, Alec Radford, Gretchen Krueger, Jong Wook Kim, Sarah Kreps, et al. Release strategies and the social impacts of language models. *arXiv preprint arXiv:1908.09203*, 2019.
- [26] Daphne Ippolito, Daniel Duckworth, Chris Callison-Burch, and Douglas Eck. Automatic detection of generated text is easiest when humans are fooled. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 1808–1822, 2020.
- [27] Sandra Mitrović, Davide Andreoletti, and Omran Ayoub. ChatGPT or human? Detect and explain. Explaining decisions of machine learning model for detecting short ChatGPT-generated text. *arXiv preprint arXiv:2301.13852*, 2023.
- [28] Hongyi Zhou, Jin Zhu, Pingfan Su, Kai Ye, Ying Yang, Shakeel AOB Gavioli-Akilagun, and Chengchun Shi. AdaDetectGPT: Adaptive detection of LLM-generated text with statistical guarantees. *arXiv preprint arXiv:2510.01268*, 2025.
- [29] Zhiguang Yang, Gejian Zhao, and Hanzhou Wu. Watermarking for large language models: A survey. *Mathematics*, 13(9):1420, 2025.
- [30] Xuhong Wang, Haoyu Jiang, Yi Yu, Jingru Yu, Yilun Lin, Ping Yi, Yingchun Wang, Yu Qiao, Li Li, and Fei-Yue Wang. Building intelligence identification system via large language model watermarking: a survey and beyond. *Artificial Intelligence Review*, 58(8):249, 2025.
- [31] Peigen Ye, Huali Ren, Zhengdao Li, Anli Yan, Hongyang Yan, Shaowei Wang, and Jin Li. Securing large language models: A survey of watermarking and fingerprinting techniques. *ACM Computing Surveys*, 58(7):1–35, 2026.
- [32] Xiao Yu, Kejiang Chen, Qi Yang, Weiming Zhang, and Nenghai Yu. Text fluoroscopy: Detecting LLM-generated text through intrinsic features. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 15838–15846, 2024.
- [33] Xin Chen, Junchao Wu, Shu Yang, Runzhe Zhan, Zeyu Wu, Ziyang Luo, Di Wang, Min Yang, Lidia S Chao, and Derek F Wong. Repreguard: Detecting LLM-generated text by revealing hidden representation patterns. *Transactions of the Association for Computational Linguistics*, 13:1812–1831, 2025.
- [34] Nadir Durrani, Hassan Sajjad, Fahim Dalvi, and Yonatan Belinkov. Analyzing individual neurons in pre-trained language models. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4865–4880, 2020.
- [35] Anna Hedström, Salim I Amoukou, Tom Bewley, Saumitra Mishra, and Manuela Veloso. To steer or not to steer? mechanistic error reduction with abstention for language models. *arXiv preprint arXiv:2510.13290*, 2025.
- [36] Elena Voita, Javier Ferrando, and Christoforos Nalmpantis. Neurons in large language models: Dead, n-gram, positional. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 1288–1301, 2024.
- [37] Asma Ghandeharioun, Avi Caciularu, Adam Pearce, Lucas Dixon, and Mor Geva. Patchscopes: a unifying framework for inspecting hidden representations of language models. In *Proceedings of the 41st International Conference on Machine Learning*, pages 15466–15490, 2024.